

EXPLORATORY STUDY ON STUDENTS' UNDERSTANDING OF MULTI-PERSPECTIVE MODELLING

Charlotte Verbruggen^{1[0000-0003-0418-2633]} and Monique Snoeck^{2[0000-0002-3824-3214]}

¹charlotte.verbruggen@kuleuven.be ²monique.snoeck@kuleuven.be

Research Centre for Information Systems Engineering, KU Leuven, Naamsestraat 69, 3000 Leuven,
Belgium

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Exploratory study on students' understanding of multi-perspective modelling

✉Charlotte Verbruggen¹[0000-0003-0418-2633] and Monique Snoeck²[0000-0002-3824-3214]

¹ charlotte.verbruggen@kuleuven.be ² monique.snoeck@kuleuven.be
Research Centre for Information Systems Engineering, KU Leuven, Naamsestraat 69, 3000
Leuven, Belgium

Abstract. Computer Science education programs often include courses on both UML and BPMN to teach students the methodology and the principles behind information systems development. While these modelling languages are typically taught in separate courses or course sections, in practice, the UML Class diagram and the BPMN process model provide different perspectives of the same information system. Therefore, students should be taught how to differentiate between these perspectives and maintain consistency between them. The goal of this exploratory study is to determine the effect that actively modelling both the UML Class diagram and the BPMN process model for a same case has on the students' understanding of multi-perspective modelling. This is done by means of an observational study with 6 students. The students were asked to think out loud while first creating a UML class diagram and then creating a BPMN process model for a given case. At the end of the study, the students filled in a questionnaire about their experience. The most important result from this experiment is that there is a potential correlation between the understanding of multi-perspective modeling and overall model quality. The study highlights points of consideration for future studies, such as time constraints and modelling experience of the participants, and prompts new research questions for further research.

Keywords: Exploratory study, UML Class diagram, BPMN process model, multi-modelling.

1 Introduction

Both UML and BPMN are frequently used in education to teach students the methodology and the principles behind information systems development. UML Class diagrams are often used for teaching database or software structure while BPMN provides the business process perspective. These languages would be typically taught in different courses or course sections, and would be illustrated with cases and exercises tailored to illustrate specific features of the language. In practice, however, the UML Class diagram and the BPMN process model provide different perspectives of the same information system. It is therefore important that students learn to differentiate between the perspectives (i.e. know what to capture in what perspective) while maintaining consistency between these two perspectives. In other words, learning the individual

languages is not enough: a skilled modeler needs to integrate the knowledge of both languages so as to be able to co-model the data and process perspective of a given process-aware information system. Although there are many courses on UML Class diagrams and BPMN process models and these languages may be taught in a single course addressing enterprise modelling (e.g. teaching ARIS, 4EM, MEMO), devoting specific attention to teaching the integration of both can instill important insights in students on the overall design of a process-aware information system.

The course “Architecture and Modelling of Management Information Systems” at the KU Leuven fulfils this need by teaching the MERODE approach, with multi-modelling being the most important learning objective. The approach is structured in three layers, each encapsulating one or more perspectives. The first layer is the enterprise layer (EL) which contains two sub-layers: the domain layer (DL) and the event handling layer (EHL). The DL consists of a UML Class diagram where each association must express existence dependency. Each Business Object in the DL is further completed with an Object Life Cycle (OLC). The events that trigger transitions in the OLCs of the Business Objects, are captured in the EHL. The EHL contains an Object Event Table (OET) that maps each business event to the Business Objects it affects. The second layer is the Information System Services Layer (ISL). This layer captures the input and output services that provide access to the EL. Finally, the third layer is the Business Process Layer (BPL) where the input and output services from the ISL can be invoked. Currently, the learning goals of the course include “Upon completion of this course, the student is capable of organizing requirements in a layered architecture”, “Upon completion of this course, the student is capable of performing a requirements analysis to create an enterprise model” and “Upon completion of this course, the student is able to relate the enterprise model and the information system services to a business process model.” In other words, students have to identify which requirements belong to the business process model, but they don’t have to create this model. The course thus sets a step towards the integration of domain modelling and BP modelling, but there is still a significant gap towards a true integrated multi-modelling approach. The goal of this paper is to investigate students’ understanding of multi-perspective modelling, more specifically, the integration of the domain model and the business process model. This is done by means of an observational study with 6 students. The goal of the observational study is to determine whether students’ understanding of which requirement is relevant for what layer is improved by modelling the business process in combination with the domain model, instead of only modelling the enterprise layer.

This paper is structured as follows. In chapter 2, we provide an overview of related work. Chapter 3 describes the methodology used for the experiment. Chapter 4 reports the results of the experiment, which are discussed in Chapter 5.

2 Related Work

When querying Google Scholar and Web of Science, we searched on the keywords (multi-modelling OR "integrated modelling" OR "integrated modeling") AND

(teaching OR education OR learning OR instruction). In web of science, we looked in all relevant categories for software and information systems modelling, and in google scholar, we filtered on articles published in the SoSyM journal. We found zero publications in in Software and Information Systems Modelling conference proceedings or journals that focus on the topic of teaching multi-perspective modelling. Although a lot of research exists on teaching modelling techniques, most papers report on teaching of an individual modelling language or perspective, not an integrated approach with multiple perspectives. Rosenthal et al. [1] provide a literature review on conceptual modelling in education. Their selection consists of 121 published papers from 1986 until 2017. The majority of papers were published after 2004. They classified the publications in distinct groups based on the modelling purpose they focus on. Object-oriented modeling is the biggest topic, with 51% of the papers focusing on this topic. The other categories identified by Rosenthal et al. are data modeling (27%), conceptual modeling in general (11%), business process modelling (8%) enterprise modelling (2%) and goal modelling with i* (1%). While Object-oriented modeling and enterprise modelling can capture multi-perspective modelling approaches, this was not discussed as an emerging research theme. The literature review provides an overview of the learning paradigms, learning approaches, learning theories and teaching methods mentioned in these publications. It is notable that not many publications discuss these aspects. However, the publications that discuss a learning paradigm mostly mention the constructivism paradigm, where “the learner is viewed as independently constructing her own subjective representations and understandings of reality through critical reflection” [1]. The publications mentioning a learning approach focus on collaborative learning. Finally, Rosenthal et al. identify four emerging research themes. The most important research theme is “Learning tool support”, especially learning support built into object-oriented modelling tools and data modelling tools. The second research theme is “Feedback”, with 22 publications reporting on process-oriented feedback and 10 publications reporting on outcome feedback. The other two research themes are “Learning Analytics” and “Gamification/Serious games”. Regarding learning analytics, Rosenthal et al. [1] observe that learning analytics in conceptual modelling education is mostly focused on datamining of logging data from modelling tools. They identify the limitation that this approach neglects other aspects of the learning process, such as “learner motivation and willingness-to-learn or the use of additional tools outside of the modelling tool, e.g., paper-based modeling”. They suggest thinking-out-loud experiments to fill this gap.

While teaching conceptual data modelling is the topic of many studies, there is still a lack of understanding what makes modelling a difficult task, and how the learning of conceptual modelling can be supported. Rosenthal et al. [2] study the difficulties that experienced modelers face in Data Modelling. They investigated the modelers’ experience using a mixed-method approach, including recording the modelers while they think aloud. They contrast this with the experiences of novice modelers. Bogdanova and Snoeck [3] present a framework for the education of conceptual data modelling based on Bloom’s taxonomy. The aim of this framework is to link the assessment of students to the related learning outcomes of the course, and to provide appropriate feedback as a means to support learning. The framework can be used to automate the process of providing personalized feedback, as demonstrated in [4], [5]. Besides feedback, also

prototyping can be a useful tool for helping the learning to understand the meaning of a model better [6]–[8]. Amongst the most frequently occurring difficulties mentioned in [2] and [9] is the modelling of relationships. Nevertheless, when a case description is intentionally cluttered with information relating to other aspects, the research in [9] demonstrates that (unless specific training is provided) up to 30% of the solutions contain superfluous classes, part of which are due to attempts to capture too many aspects in a single perspective. A sequel research mapping errors to learning objectives [4], reveals that out the four errors appearing in more than 60% of the tasks, three are related to the learning objective of distinguishing between EL, IS and BP layer requirements.

Process modelling is also a prevalent topic in Computer Science education, as indicated by the large amount of studies done about the cognitive aspects of process modelling. For example, Figl et al. [10] analyzed the influence of notational aspects on the comprehension of process models. In a different study, Figl and Laue [11] looked at the comprehensive complexity of relationships between elements in a process model. Burattin et al. [12] developed a machine learning approach to automatically identify the different phases a modeler goes through when creating a process model. In the area of process modelling, the integration of full-fledged data modelling is still under development [13], and research analyzing modelling errors related to a multi-modelling has not been found.

When considering research on students understanding of modelling languages or on recurring difficulties in teaching modelling, all of the afore-mentioned research papers address only one modelling language (e.g., BPMN, UML Class diagrams, ER, etc.). To the authors' best knowledge, research that addresses the learning or teaching of a multi-modelling approach is inexistent. The goal of the research presented in this paper is to study to what extent the co-modelling of different perspectives of a single case study can shed light on the problems faced by students, in view of creating better support for learning multi-modelling.

3 Methodology

The methodology used in this paper is inspired by [2] and aims to capture the cognitive process of novice modelers by means of a thinking-out-loud experiment. The aim of the experiment is to determine whether modelling the business process in combination with the domain model, improves the understanding of the layered structure of MERODE, compared to just modelling the enterprise layer.

In the experiment, a number of students are given a case description (see Appendix 2) and asked to model the enterprise layer as a UML class diagram and the business process layer as a BPMN process model (see Appendix 1 for model solutions). The selected case description for this exercise is adapted from [14], as that case lends itself to multi-perspective modelling: it contains ingredients for both a process model and a data model and the ingredients are not separated. So, the students have to filter out themselves what is relevant for the process model and what is relevant for the data model.

The participants are all students in the course “Architecture and Modelling of Management Information Systems”. At the time of the experiment, the participants have just finished week 3 of the course, meaning they were introduced to the layered approach of MERODE and had seen chapter 1 to 3 of the handbook “Enterprise Information Systems Engineering” [15]. The students following this course also have had a prior course on database management, and most followed at least one course where the BPMN notation is taught. The call for participation included the prerequisite that participants should have some experience with UML class diagrams and BPMN process models. Before starting the experiment, the participants were sent a short demo of the tool Signavio.

3.1 Experiment Set-Up

Based on the recommendation by Rosenthal et al. [2], we use a multi-modal approach, combining several

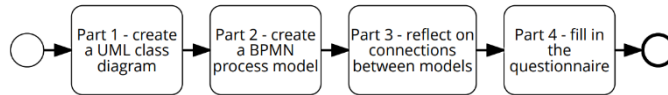


Fig. 1. Experiment process

methods of data collection. The cognitive processes of the participants are recorded in various ways in order to capture as much information as possible. Participants are asked to think out loud which is audio-recorded, their modelling actions are captured via screen recording, and an over-the-shoulder video-recording captures the annotations they make to the case description provided on paper. The experiment consists of 4 parts, as shown in Figure 1. In the first part, the participants create a domain model for the given case by means of a UML class diagram. Then, they model the process for the same case in a BPMN process model. In the third part, they indicate at which places in the BPMN model it is necessary to access the information from the UML class diagram. Finally, they are asked to fill a questionnaire on their experience during the experiment.

4 Results

4.1 Measured performance

In first instance, we analyzed the models created by the participants by listing the mistakes found in the models. Each mistake is grouped into a mistake type (e.g. “missing task”). These mistake types are then categorized again into one of four mistake groups: “BPMN model”, “UML class diagram”, “layer allocation”, and “notation”. The “layer allocation” category consists of mistakes where a certain requirement was accounted for in the wrong model. For example, when “HR Department” was modelled as a class in the UML class diagram, while it is the user of the system, and thus needs (only) to appear as actor in the BPMN diagram. The category of “notation” deals with notational issues (i.e. syntactical issues), either for BPMN models or for UML class diagrams, for

example, an exclusive-or gateway in the BPMN model that has only one incoming flow and only one outgoing flow.

Table 1. Overview of mistakes made by the 6 participants

	Participant						Total	Total %
	1	2	3	4	5	6		
BPMN model	12	10	8	7	16	16	69	55,2%
missing path	4	1	1	1	5	3	15	
missing task	4	2	2	2	6	7	23	
missing timer	2	2		1	2		7	
multiplicity of task	1	1			1	1	4	
Universe of discourse as pool		1					1	
overspecification of task	1	1	1	1	1	1	6	
task in wrong pool		2	2			1	5	
missing pool			2	1	1	1	5	
unneeded subprocess				1		1	2	
unneeded pool						1	1	
layer allocation	3	4	2	1	0	3	13	10,4%
attribute as task		1					1	
actor as class	2	3				3	8	
class as task	1		1	1			3	
input form as class			1				1	
notation	5	5	3	5	1	4	23	18,4%
BPMN - event is not atomic	1	2					3	
BPMN - loose task	1					1	2	
BPMN - missing end event	2	1				1	4	
UML - association as attribute			1	1			2	
BPMN - missing message flows		1		1		1	3	
BPMN - pool name missing						1	1	
BPMN - unneeded gateway				1	1		2	
UML - missing multiplicity			2	2			4	
UML - unconnected diagram	1	1					2	
UML class diagram	3	6	1	3	3	4	20	16,0%
wrong attributes		1					1	
association multiplicity		2				2	4	
missing association		1					1	
missing class	2	1	1	1	2	2	9	
unneeded class	1			1	1		3	
Universe of discourse as class		1					1	
class as attribute				1			1	
Total	23	25	14	16	20	27	125	100,0%

In total, the models contain 125 mistakes, both syntactic and semantic (see Table 1). More than half of the mistakes were found in the BPMN models. The UML class diagrams account for 16% of the mistakes, and 18,4% of the mistakes are related to notation. The smallest group are “layer allocation” mistakes, accounting for 10,4% of the

mistakes. However, they might have repercussions on the quality of the model that represent the distinct perspectives. Therefore, we discuss them in more detail below.

The most frequently occurring error in the UML class diagram is the missing class. Of the four classes that were expected (Job Vacancy – Candidate/Application – Review – Interview), Candidate was the only class present in all solutions. Job Vacancy was part of three solutions. Review and Interview each appeared only once in a different solution, even though we made sure to include sentences in the case description that suggest modelling these as classes: “All reviews [...] should be filed within four weeks of being requested” and “[...] interviews are registered in the system”.

The most frequently occurring errors in the BPMN model are the missing task and the missing path. Often this is due to an incomplete model, see the discussion section.

The group of “layer allocation” mistakes consists of four types of mistakes. The mistake type that occurs most often is “**actor as UML class**”, meaning that, in the scope of a small application, an actor such as “HR Department”, “Professor” or “International Office” is unnecessarily modelled as a class in the UML class diagram. This mistake is made by three out of six participants and each of those participants makes this mistake multiple times. It is notable that in seven out of eight cases where this mistake is made, the actor is modeled both as a class in the UML class diagram and as a pool in the BPMN model. In the other case, the participant ran out of time. The second mistake type is “**class as task**”, meaning that the BPMN model contains tasks like “review the candidate” and “register interview”, while the corresponding classes where the information is stored are missing from the UML class diagram. This mistake occurs once in three different solutions. The other two mistake types occur only once each: “**attribute as task**” and “**input form as class**”. On average, participants made two layer allocation mistakes and only one participant did not make any layer allocation mistakes.

Our sample of six participants is too small to make a statistical analysis, but overall, the ratio of layer allocation mistakes to total mistakes seems fairly consistent around 13% (not counting the participant that didn’t make any layer allocation mistakes).

4.2 Recordings

We can inspect the time spent on each part of the exercise. Each participant was given a timeslot of one hour to work on the exercise and fill in the survey. However, four of the six participants were unable to finish the exercise in the allocated time. This explains that 58% of the mistakes are related to missing elements. Noticeably, the two participants that did not run out of time, spent about 30% of their time on the UML class diagram, while the other four participants spent 50%-60% on the UML class diagram (see Table 2). Those two ‘faster’ participants have a low number of layer allocation mistakes. They also have the lowest ratio of total mistakes to total number of elements and they spent noticeably more time explaining the connection between the models. They are the only two students that made changes to their models in the third part of the exercise.

Table 2. Division of time over models

Participant	1	2	3	4	5	6
% time UML	52%	50%	29%	34%	52%	59%
% time BPMN	42%	49%	57%	55%	44%	35%
% time explaining connections	6%	1%	15%	12%	5%	6%

By analyzing the video recordings, we can conclude that the three participants that have the least amount of layer allocation mistakes (0, 1 or 2) were able to connect their models to each other to some extent. The other three participants made more mistakes (3 or 4) and did not understand how to connect their models.

Finally, we look at the type of notes participants make on the case description. Participants were given a pen and a set of markers. All participants highlighted words (this includes underlining or circling in pen) while reading the case for the first time. Only three participants continued making notes, highlights or sketches while they were modelling. We did not find any meaningful correlations between note taking and the understanding of multi-perspective modelling.

4.3 Questionnaire

The first two questions of the questionnaire are about previous experience with UML class diagrams and BPMN models. Half of the participants (1, 2, 3) indicate that they have some experience with UML class diagrams, and half indicate that they have a reasonable amount of experience (e.g. active participation in a course). For BPMN, participant 2 has some experience, the others indicate having a reasonable amount.

The questionnaire also investigates their perceived understanding of the layers during the creation of the UML model, and how this evolved over the next parts of the exercise. Half of the participants (3, 4, 6) indicate that they have some issues allocating requirements to layers during the creation of the UML class diagram. The only participant experiencing no layer allocation problems (5) indeed did not include superfluous classes in the UML diagram. However, the process model was highly incomplete. Half of the participants (1, 3, 4) also agree that creating the BPMN model helped them to identify to which layer each requirement belongs. Two of these (3, 4) are the faster modelers, who also made the lowest number of errors overall. In a follow-up question, the participants can clarify how creating the BPMN model helped or didn't help. Of the three participants that don't agree that the BPMN model is helpful, two explain that they didn't think about the layers while solving the exercise, and one only discusses the BPMN model instead of the connections between both models. Those three participants (2, 5 and 6) have the highest numbers of errors overall. All participants who agree that the creation of the BPMN model was helpful for layer allocation also agree that the third part of the exercise (thinking about the connections between the models) is helpful to understand the interaction between the layers. In a follow-up question, the others indicate that they either didn't see the relation, didn't think of the layers while modelling, or have a lack of experience. All six participants agree that they have a better understanding of the case after finishing part 2 and part 3 of the exercise, and four out of six (1, 3, 4, 5) indicate that they have a better understanding of the layers in general.

The final two questions of the questionnaire investigate whether students think additional similar exercises and code generation for the business process layer would be helpful when studying the course “Architecture and Modelling of Management Information Systems”. The same group of participants that indicate that part 2 and part 3 of the exercise were helpful (1, 3, 4), agree that including this type of exercise in the course would be beneficial. Five out of six participants (all but 4) believe that code generation for the business process layer would help them.

5 Discussion

This exploratory study is a first step in investigating the use of multi-perspective modelling in computer science education. Although the experiment is small-scale, the results already indicate that more research in this domain could lead to some interesting insights. In this study, the data was gathered at the start of the course “Architecture and Modelling of Management Information Systems”. As the students progress through the course, we will be able to gather more data and perform a more in-depth analysis.

5.1 Analysis of mistakes

The type of mistakes that occurs the most (23 times in total) is a missing task in the BPMN model. In each solution, at least two tasks were missing, but generally speaking many more were missing, leading also to missing paths (15), etc. The third type of mistake is a missing class in the UML class diagram. This occurred at least once in each solution and nine times in total. The high number of missing tasks can be explained by the fact that four out of six participants spent quite some time on creating the UML class diagram and then ran out of time for modelling the business process. However, while the participants were instructed to start with the UML class diagram, they were free to decide themselves when the diagram was finished. Therefore, the time constraint does not explain the missing classes. The participants' inability to identify review and interview as potential classes, and not even including these as attributes could be explained by the participants' limited experience with data modelling.

The fourth type of mistake that occurred often, was a layer allocation mistake: “actor as class”. Three participants (3, 4, 5) did not make this mistake, the other three made the mistake multiple times. We can make two observations concerning this mistake. First, the three participants that made this mistake were also unable to reflect on the connections between both models in part 3. Two of these participants said that they didn't know what the connections were and one participant only talked about the BPMN process model. This could indicate that the interaction between the layers is a complex concept to grasp for students. Second, two out of the three students that did not make this mistake (3, 4), made less mistakes overall. Even when we discard the layer allocation mistakes, these two students made the least number of mistakes. This could indicate that once students understand the interactions between layers, the overall quality of their models improves. Participant 5 also didn't make this mistake, however, the process model was highly incomplete, thus leading to a large number of mistakes

overall. The other layer allocation mistakes were less frequent, showing that, at least in this case, the modelling of actors in a system is one of the main challenges of layer allocation. For another case this could turn out to be different.

In this analysis, we did not consider the amount of correctly modelled aspects of the case. This also influences the quality of a solution. For example, participant 2 made the most mistakes, however, their models were more complete than for example the models of participant 5. The ratio of mistakes to total number of elements in the models could account for this discrepancy. As shown in Table 3, this ratio suggests that the solution of participant 2 is better than the solutions of participants 1, 5 and 6, despite having the highest number of mistakes.

Table 3. Total mistakes vs Total number of elements

Participant	1	2	3	4	5	6	Total
total mistakes	23	25	14	16	20	27	125
total number of elements	20	50	36	37	21	21	185
Total mistakes / Total number of elements	1,15	0,50	0,39	0,43	0,95	1,29	0,68

5.2 Analysis of recordings

As indicated in the results, there seems to be a correlation between the number of layer allocation mistakes and the time spent on the UML class diagram, at least for the better performing participants 3 and 4. It would be interesting to study whether there is a causal relation between the understanding of the interaction between layers and ease of identifying classes for the UML class diagram, reducing the time spent on this task. Alternatively, spending more time on the business process layer could improve the understanding of the interaction between layers, resulting in less mistakes.

Another interesting observation from the recordings is that one of the participants who understands the interaction between the layers well, explicitly mentioned that they would prefer to switch between the models and not follow the instructed order. This was the only participant who explicitly modelled the connections between the models by including in the BPMN model datastores representing the classes of the UML class diagram. Repeating the experiment where students are allowed to determine the order themselves, could provide more insight into the cognitive process of novice modelers.

5.3 Analysis of survey results

A first observation regarding the questionnaire, is that the level of experience in UML class diagrams was lower than expected. Since the participants were all students of the program “Master of Information Management” or “Master of Business and Information Systems Engineering”, we assume they have all taken the courses “Business Information Systems” and “Principles of Database Management”, as these courses are mandatory in both programs that are planned before the “Architecture and Modelling of Management Information Systems” course. Both courses include UML class diagrams. However, three participants indicated they only have some experience with

UML class diagrams, instead of choosing the option “*I have a reasonable amount of experience (e.g. active participation in a course about UML class diagrams)*”.

The second observation is that there is a slight discrepancy between the answers to some questions. More specifically, only half of the participants indicated that Part 2 and 3 were helpful for understanding the interaction between the layers, but all participants agreed that Part 2 and Part 3 contributed to their overall understanding of the case. Apparently, understanding the case is different from understanding the different layers and perspectives present in the case description. This could indicate a mismatch between the actual issues novice modelers have (understanding to which perspective a requirement relates), and their perception of the issues that they face (understanding a requirement).

Finally, our third observation is related to the last two questions, which measure the perceived usefulness of these types of exercises and prototyping of the business process layer. While not all participants agreed that these would be useful additions to the course material, no participant showed disagreement, instead choosing for the option “Neutral”. So overall, we can expect a positive response to these additions.

5.4 Reflection on methodology

In this study, we used a multi-modal data collection method in order to capture as much information as possible. Out of the different modes that were used for the collection of data, the correction of the participants’ solutions, the screen and voice recordings and the questionnaire yielded interesting information. However, the recordings of the notation did not bring any additional insights. To simplify the experiment, we might consider leaving this out. In view of repeating the experiment with larger groups, and considering the amount of time needed for processing all the data, in future repetitions of this study we will consider focusing on correcting mistakes, data gathering while observing and a well-designed survey. Data gathering while observing would significantly reduce the time spent on transcribing recordings, since the recordings would serve mostly verification for the gathered data and not as a primary source.

5.5 Limitations

The exploratory nature of this study comes with a number of limitations. The first limitation is the small number of participants. With only six participants, the results of this study are only an indication of the understanding that novice modelers have of multi-perspective modelling. Nevertheless, the insights gathered in terms of timing, order of tasks, etc., provides useful insights in how to set up experiments with larger groups of participants to obtain more significant results.

As the participants come from an international group of students, none of them were native English speakers, and they had differing levels of fluency in the English language. Also, not all participants were accustomed to thinking-out-loud exercises. This likely led to a loss of information about their cognitive process. Despite this loss of information, some interesting insights could already be captured from this observational study.

A third limitation, as indicated in the previous section of the discussion, is that some participants had limited experience with UML class diagrams. Also, some participants seemed to have limited experience with Signavio, despite receiving a demo beforehand and using this tool in a previous course. This probably led to the fact that most participants needed more time than we anticipated, resulting in incomplete BPMN diagrams. For future experiments, we can consider providing tool training beforehand.

A fourth limitation is that the students were only given one case. It would be interesting to investigate if the results of this experiment persist when students are given different cases. That way, we could eliminate the wording and specific aspects of a given case as cause for issues.

A fifth limitation of the study is that there is always room for interpretation when creating a model based on a case description. This was taken into account when correcting and comparing the solutions in the following way: we didn't penalize for sub-optimal naming of model elements, missing conditions on gateway paths or missing attributes. To further mitigate for this problem, we use a case from another researcher, for which we created an own solution that turned out to be almost identical to the original solution. Both the original solution [14] and our own variant [13] were published in the context of papers that underwent a review process. In this sense, we can consider the case and its solution as validated by several experts.

6 Conclusion

This exploratory study is a first step in the investigation of novice modelers' understanding of multi-perspective modeling. The study uses a combination of participant's solutions to the exercise, recordings and a survey to gather data. The inclusion of the recordings and the survey added meaning to the overview of mistakes made. The study shows promising results including a potential correlation between the understanding of multi-perspective modeling and overall model quality, and between the understanding of multi-perspective modeling and the time distribution over the two models. Further research is needed to confirm these findings and to investigate if there is a causal relation. Another result is that the modelling of actors in the system can be a challenge for novice modelers. The study highlights points of consideration for future studies, such as time constraints and modelling experience of the participants, and prompts new research questions for further research.

Appendix 1 – Model Solution

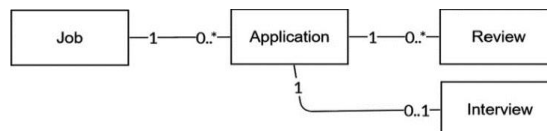


Fig. 2. UML class diagram - model solution

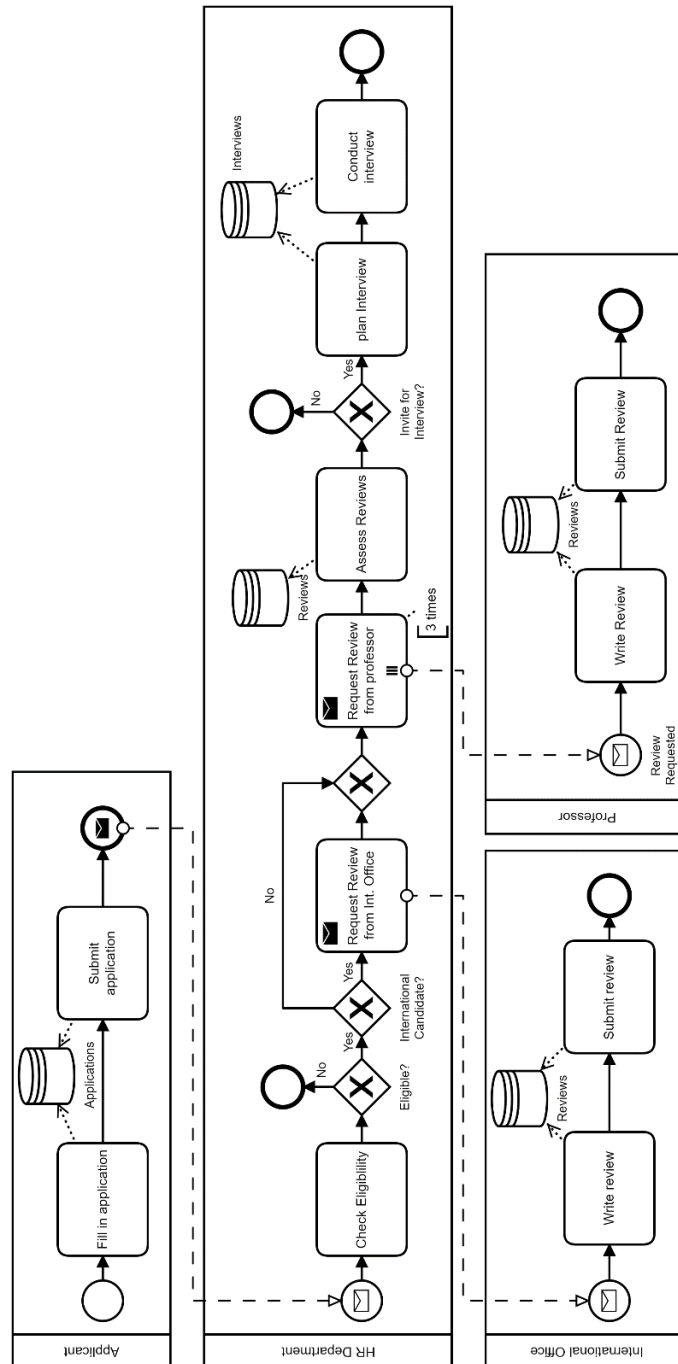


Fig. 3. BPMN process model - model solution

Appendix 2 – Case Description

The KU Leuven wants to develop a web application for the recruitment of new PhD candidates. In this exercise, you will model the part of the system that deals with entering and reviewing new applications. The creation of a new vacancy and deciding which candidate to hire are out of scope for this exercise. The requirements are stated below:

In order to apply for a job vacancy, candidates have to fill in a form where they specify their personal information (name, surname, e-mail address, birthday, nationality) and they upload their grades transcript and motivation letter. The application can be saved and submitted when the applicant is ready. The application should be submitted before the deadline specified in the vacancy. Once the application is submitted, the HR department is notified and the application is assigned to an HR officer. After submission, the candidate can no longer make changes to the application.

The HR department will then make a first assessment of the application. They decide whether or not an application is eligible based on the obtained degree, grades transcript, university ranking, language certificates, and GMAT or GRE-score. If an application is ineligible, the candidate is immediately rejected. If an application is eligible, it must be reviewed by several people. First, international candidates will be reviewed by the international office. Once the international office has written a review, the HR department will then contact three professors with the request to write a review of the application. All reviews, both from the international office and from professors, should be filed within four weeks of being requested. Each review concludes with a proposal for the next step in the recruitment process: rejection, or an interview. When the deadline of the vacancy is passed, the HR office will decide which candidates to invite for an interview based on the reviews. Once a date has been set, interviews are registered in the system. The interviewers will fill in a form with their comments & conclusions.

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